



Evaluating climate change impact on drought: a comprehensive review of drought indices and future projections

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Abstract

Drought, a complex and insidious natural hazard, poses a significant and escalating threat to global water security, agricultural production, ecosystems, and human livelihoods. Unlike sudden-onset disasters, droughts are “creeping phenomena,” slowly developing periods of below-average precipitation and reduced water availability that can persist for months to years, with cascading socio-economic consequences. In an era profoundly shaped by anthropogenic climate change, understanding and predicting drought dynamics are more critical than ever. Climate change exacerbates drought conditions, increasing their frequency, intensity, and duration, though impacts vary across regions. Key drought indices, including the Standardised Precipitation Index (SPI), the Standardised Precipitation Evapotranspiration Index (SPEI), and the Palmer Drought Severity Index (PDSI), are crucial for assessing drought severity. Advances in remote sensing and multi-variable indices have significantly improved monitoring accuracy, particularly for agricultural droughts. The application of artificial intelligence (AI) and machine learning (ML) has enhanced drought prediction by integrating multiple climate variables, though challenges related to data reliability and model transparency remain. High-resolution datasets, such as daily SPI/SPEI and weekly drought indices, contribute to more precise monitoring and forecasting. Future projections indicate an increase in agricultural and hydrological droughts, highlighting the importance of multi-index assessments and advanced climate models for improved forecasting and resource management. Ongoing developments in drought indices, AI-driven analysis, and high-resolution monitoring are crucial for strengthening drought mitigation efforts, supporting climate adaptation, and ensuring effective water resource planning.

Keywords Drought · Climate change · Machine learning (ML) · Artificial intelligence (AI) · Climate models · Drought mitigation

1 Introduction

1.1 Definition of drought

Drought, typically defined as a prolonged period of below-average precipitation, remains one of the most complex and least understood natural hazards due to its slow onset, diverse manifestations, and far-reaching impacts on ecosystems, agriculture, and societies (Wilhite and Glantz 1985). With the intensification of anthropogenic climate change, droughts have become more frequent, prolonged, and severe across many parts of the world (Ghazi et al. 2025), underscoring the urgent need for robust monitoring and assessment frameworks. Unlike rapid-onset disasters such as floods or storms, droughts evolve gradually and can persist for months or years, making their detection and response particularly challenging (Zhai et al. 2021). Over the past few decades, significant progress has been made in conceptualising and categorising drought into meteorological, agricultural, hydrological, and socioeconomic types (Palmer 1965; Dai 2011; Van Loon 2015; Wilhite 2021). This lack of consensus is further complicated by the proliferation of indices ranging from climatic (e.g., SPI, PDSI) to remote sensing-based (e.g., VCI, TCI), and composite indicators (e.g., SPEI, CDI), each designed to capture different dimensions of drought (Choukri et al. 2023). Recent reviews, including Mukherjee et al. (2018), have emphasised the need for comparative evaluations of these indices across diverse climatic regions to improve our understanding of their performance and suitability. Emerging research has also highlighted new drought phenomena such as “flash droughts”, rapid-onset events that have gained attention due to their severe and unexpected impacts (Lisonbee et al. 2022). These developments highlight the increasing complexity of drought as a dynamic, process-oriented phenomenon shaped by feedbacks among climate, land use, and human systems (AghaKouchak et al. 2021).

Despite these advances, critical knowledge gaps remain regarding how different drought indices perform across diverse geographic and climatic contexts. While individual studies often evaluate specific indices in isolated regions, there is a lack of comprehensive, cross-regional assessments that consider multiple indices using consistent metrics. Addressing this gap is crucial for enhancing drought monitoring, forecasting, and management strategies in a changing climate. The primary objective of this study is to systematically evaluate and compare the average performance of key drought indices, including SPI, SPEI, PDSI, CDI, VCI, TCI, and SPERI across ten major global regions. By synthesising performance metrics, this research aims to (1) identify which indices perform consistently well or poorly across different environments, (2) highlight region-specific strengths and limitations of each index, and (3) provide guidance for selecting appropriate indices for future drought risk assessments and adaptation planning.

1.2 Importance of studying drought in the context of climate change

Understanding drought in the context of climate change is critical due to its far-reaching effects on water security, agriculture, ecosystems, and socioeconomic stability. As global temperatures rise and precipitation patterns become more erratic, drought events are projected to increase in frequency, intensity, and duration, thereby amplifying existing vulnerabilities and stressing both natural and human systems (Nam et al. 2015; Lesk et al. 2021;

Han and Singh 2023). These emerging trends underscore the need for accurate, regionally relevant drought assessments to support proactive planning and climate-resilient adaptation strategies (Kim and Jehanzaib 2020). However, the impacts of climate change on drought are not uniform worldwide. While some regions are projected to face more frequent and severe droughts, others may experience less frequent or even shifting seasonal patterns (Kim et al. 2020). Drought monitoring relies on a wide array of indices, including meteorological (e.g., SPI, PDSI), agricultural (e.g., VCI, TCI), and composite indices (e.g., SPEI, CDI). However, no single index is universally applicable due to differences in data requirements, temporal resolution, and sensitivity to specific drought characteristics (Mukherjee et al. 2018; Choukri et al. 2023). Mukherjee et al. (2018) provided an important synthesis of drought indices and their relevance under climate change; however, significant gaps remain, particularly in terms of comparing the performance of these indices across multiple global regions under consistent evaluation frameworks. Moreover, recent literature has called for improved methods that address the non-stationarity of climate processes, the role of anthropogenic influences, and the complex feedbacks between climate, land use, and water resources (AghaKouchak et al. 2021; Han and Singh 2023). These demands have led to growing interest in developing and evaluating indices that integrate diverse data sources (e.g., precipitation, evapotranspiration, vegetation health) to provide a more comprehensive view of drought risk.

1.3 A brief overview of drought indices

Drought indices play a crucial role in monitoring and quantifying drought severity, providing essential tools for water resource management, agricultural planning, and assessing climate impacts. These indices have evolved significantly from simple precipitation-based metrics to more complex models that integrate temperature, evapotranspiration, soil moisture, and vegetation health (Heim 2002). Among the most widely used indices are the Standardised Precipitation Index (SPI), suitable for precipitation-driven droughts in regions like Western and Central Europe (Tigkas et al. 2015; Raible et al. 2017), and the Palmer Drought Severity Index (PDSI), which incorporates temperature and soil moisture balance and has seen extensive application since the 1960s (Tian et al. 2018). Recent advancements have introduced indices like the Standardised Precipitation Evapotranspiration Index (SPEI), which accounts for the increasing role of temperature in drought dynamics, especially under climate change, and is considered particularly relevant for regions such as the Mediterranean (Raible et al. 2017; Tian et al. 2018). Other indices, such as the Reconnaissance Drought Index (RDI) and Streamflow Drought Index (SDI), have also been developed to enhance flexibility and hydrological relevance (Tigkas et al. 2015). Seminal reviews, such as those by Mukherjee et al. (2018), have critically evaluated the suitability of drought indices under climate change, emphasising the need for indices that incorporate non-stationarity, account for anthropogenic influences, and offer cross-sectoral applicability. Despite these insights, significant knowledge gaps remain, particularly in terms of systematically comparing the performance of multiple drought indices across diverse geographic and climatic regions using unified metrics. Existing studies often focus on specific regions or indices, limiting their utility for broader-scale drought monitoring and adaptation planning. This study addresses these gaps by conducting a comprehensive comparative assessment of seven widely used drought indices, SPI, SPEI, PDSI, RDI, CDI, TCI, and VCI across ten global regions.

1.4 Overview of climate change

Climate change, driven largely by anthropogenic activities such as fossil fuel combustion, deforestation, and industrialisation, is fundamentally altering global weather systems (VenkataRaman et al. 2012; Byg and Shah 2024). Its consequences, including rising temperatures, shifting precipitation patterns, and an increased frequency of extreme events, pose multifaceted risks to ecosystems, water resources, and human well-being (Weiskopf et al. 2020; Zhao et al. 2022b). Among these, drought stands out as a particularly complex and consequential hazard due to its creeping nature, prolonged duration, and broad socio-environmental impacts. Climate change intensifies droughts by exacerbating evapotranspiration and disrupting hydrological cycles, thereby threatening food and water security and undermining sustainable development goals (Levy et al. 2017; Walker 2018). Some regions may experience worsening drought conditions, while others might see changing seasonal drought patterns or altered intensity (Zhao et al. 2022a). This spatial heterogeneity calls for region-specific monitoring tools and robust analytical frameworks. One of the most critical challenges lies in how we measure, monitor, and predict drought under evolving climatic conditions. Drought indices tools, designed to quantify drought based on variables such as precipitation, temperature, soil moisture, and vegetation, are essential to this task. However, these indices differ widely in structure, sensitivity, and applicability, and no single index can universally capture drought characteristics across all regions and time scales.

Seminal works, such as Mukherjee et al. (2018), have emphasised the need to update and evaluate drought indices in light of climate change, particularly with respect to their ability to account for non-stationarity, human influences, and cross-sectoral impacts. Despite such efforts, comparative assessments of multiple drought indices across diverse geographic regions using standardised performance criteria remain scarce. Moreover, most existing studies focus narrowly on specific indices or regional case studies, limiting their generalizability and practical value for global drought risk management. Figure 1 illustrates the

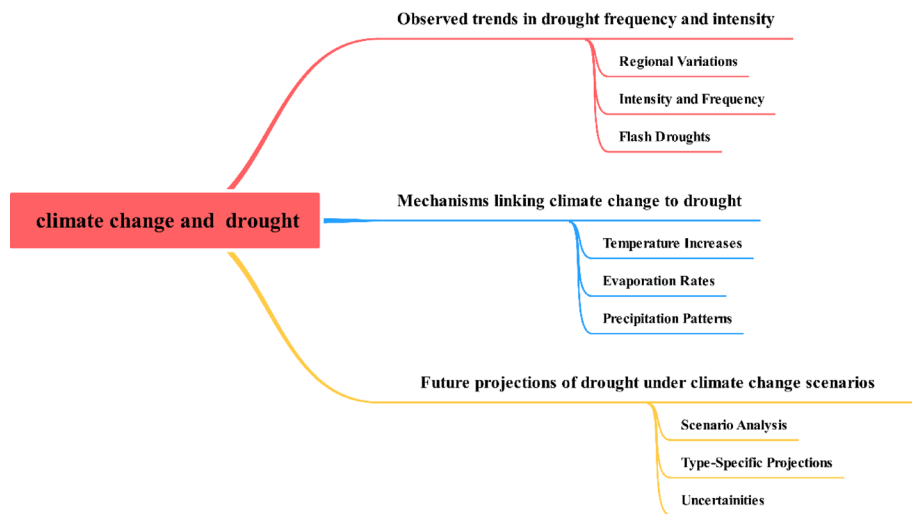


Fig. 1 Analysis of climate change impacts on drought: mechanisms, historical trends, and future projections

conceptual framework linking climate change and drought, encompassing observed trends, underlying mechanisms, and future projections.

1.5 Mechanisms linking climate change to drought

Rising temperatures increase evaporation rates and atmospheric water demand, exacerbating drought conditions (Zhang et al. 2014; Tadese et al. 2020). The “upped-ante mechanism” describes how warming increases the moisture threshold required for convection, potentially reducing rainfall in marginal convection zones (Neelin et al. 2003). Climate models project significant regional changes in tropical rainfall, with some areas experiencing increased drought risk (Neelin et al. 2003; Cammalleri et al. 2020). In Europe, drought severity, duration, and frequency are expected to increase in Mediterranean regions but decrease in Boreal areas as global warming progresses (Cammalleri et al. 2020). The alternation between wet and dry spells is also becoming more pronounced, thereby challenging the ability of soil microbial communities to adapt (Li et al. 2023). Climate change is significantly impacting drought patterns in Africa and South America through various mechanisms. In Africa, projections indicate a shift towards drier conditions in many regions, with South Africa and West Africa expected to experience more severe and frequent droughts by the 2050s and 2080s (Gizaw and Gan 2017). Ethiopia, however, exhibits a contrasting trend with a decrease in drought occurrence under future climate scenarios (Mengistu et al. 2025). Interestingly, the link between climate anomalies and violent conflict, often attributed to drought-induced agricultural shocks, is not as straightforward as previously thought. A study comparing half a century of data found only weak and inconsistent connections between agricultural output and violent conflict in Africa, suggesting that socioeconomic and political contexts play a more significant role (Buhaug et al. 2015). In South America, climate change is expected to alter precipitation patterns and increase temperatures, potentially exacerbating drought conditions. However, the impacts may vary regionally, with some areas experiencing increased rainfall while others face more severe water scarcity (Sylla et al. 2016; Mengistu et al. 2025). The El Niño–Southern Oscillation phenomenon, which is expected to increase in frequency and intensity due to climate change, has been linked to drought outbreaks in both Africa and South America (Githeko et al. 2000). Climate change is intensifying drought conditions in the Amazon region through several interconnected mechanisms. Higher temperatures increase evaporation rates, reduce soil moisture, and alter precipitation patterns, leading to more frequent and severe droughts (Braz-Mota and Luis Val 2024). This is particularly evident in sensitive ecosystems, such as the Amazon rainforest, where reduced rainfall and higher evaporation rates result in significantly lower water levels, threatening biodiversity and human livelihoods (Braz-Mota and Luis Val 2024). Interestingly, land cover change and climate variability interact to influence the water cycle in the Amazon basin. Deforestation can decrease dry season latent heat flux, which is further exacerbated by drought conditions. During drought years, there is also an increase in dry-season moisture recycling over the Amazon Basin by ~7%, potentially reducing precipitation and amplifying drought effects (Bagley et al. 2014). This feedback loop between land cover change and climate can push the Amazon closer to a critical threshold of rainforest dieback (Boulton et al. 2022). In conclusion, the mechanisms linking climate change to drought in the Amazon region are complex and interconnected. They involve increased temperatures, altered precipitation patterns, changes in evaporation rates, and modifications to land cover. These factors com-

bine to create a potentially devastating cycle of increasing drought severity and frequency, threatening the resilience of the Amazon rainforest and its crucial role in regional and global climate regulation (Malhi 2009; Boulton et al. 2022). Gaining a deeper understanding of these processes is essential for developing effective drought management strategies and adaptation measures in response to a changing climate (Falkenmark 2013).

1.6 Observed trends in drought frequency and intensity

Flash and long-term droughts have been increasingly frequent, prolonged, and intense across many regions. From 1961 to 2016, flash drought characteristics in China showed significant upward trends, with climate warming and vegetation greening accounting for 89% and 54% of the increased frequency, respectively (Zhang et al. 2022). In Iran, the severity and extent of regional droughts have increased, particularly in the central desert basins and southeastern regions, since 2000 (Hosseini et al. 2021). In India, there has been a notable rise in the intensity and area affected by moderate droughts in recent decades, along with a statistically significant increase in the frequency of multi-year droughts (Niranjan Kumar et al. 2013). Studies in Europe also report an increasing severity of summer droughts, although the overall frequency of dry months has not shown a statistically significant trend (Dukat et al. 2022). Interestingly, some regions display contrasting trends. For example, precipitation in North America's Great Plains has decreased significantly in the Southern Plains while increasing in the Northern Plains, linked to shifts in the Great Plain's low-level jet (Barandiaran et al. 2013). In Spain, northern regions with higher flash drought frequency showed negative trends, while regions with fewer flash drought events generally exhibited positive trends (Vicente-Serrano et al. 2023). The Continental U.S. exhibited persistent drought patterns in the western and eastern regions, as well as the Great Plains, with increasing trends in drought characteristics, while the interior experienced short-term variations (Ge et al. 2016). In Africa, the frequency of droughts has increased significantly across most of the continent, particularly in equatorial regions and the Sahel (Řehoř et al. 2023; Ogunrinde et al. 2025). The Mediterranean and Sahara regions have shown the most pronounced trends towards drier conditions (Ogunrinde et al. 2025). Sub-Saharan Africa has been identified as especially vulnerable, with drought frequency expected to increase further due to climate change (Lottering et al. 2021). Analysis of the Standardised Precipitation Evapotranspiration Index (SPEI) revealed that drought severity and frequency were more pronounced in South Africa's Free State province compared to the North West province from 1985 to 2015 (Botai et al. 2016). South America has experienced some of the most dramatic increases in drought occurrence globally. More than half of the continent exhibited significant drying trends from 1981 to 2021, accompanied by an increasing frequency of extreme drought events (Řehoř et al. 2023). The Amazon Forest and southern Chile have been identified as drought hotspots in recent decades (Spinoni et al. 2014). Climate projections indicate substantial reductions in precipitation over most of South America in the coming decades, particularly during austral spring, with an increased duration and severity of drought events expected (Ferreira et al. 2023).

1.7 Future projections of drought under climate change scenarios

Future drought projections under climate change scenarios suggest complex and region-specific patterns across different drought types. Agricultural and hydrological droughts

are expected to become more frequent as global temperatures rise, while meteorological droughts may decrease in certain areas (Zhang et al. 2017).

Additionally, changes in drought intensity are expected to increase linearly with rising global temperatures, particularly under high-emission scenarios. Several studies highlight distinct responses of different drought types to global warming, with asymmetric impacts resulting in stronger effects on extreme droughts than mean conditions (Kang and Sridhar 2017; Zhang et al. 2017). This apparent contradiction emphasises the importance of considering multiple factors beyond precipitation when assessing future drought risks. Various drought indices and hydrological models yield differing projections; for instance, soil moisture-based indices suggest an increase in agricultural droughts, while other indices indicate a decrease in drought occurrences due to increased precipitation (Kang and Sridhar 2017). In conclusion, future drought projections vary significantly depending on the region, climate scenario, and drought definition used. Many studies project an increase in drought severity and frequency in arid and semi-arid regions (Shirmohammadi et al. 2013; Wu et al. 2024). However, uncertainties remain high due to variations in climate models and drought indices. Using multiple modelling approaches and drought indices that combine precipitation and temperature effects is crucial for a comprehensive assessment of drought risk under climate change (Abiodun et al. 2019).

Figure 2 provides a conceptual overview of the review's scope. It visually maps the major components addressed in the study, including drought types, causal factors, key indices, definitions, and the impacts of climate change. This framework guides the structure of the review and highlights the interconnectedness of these elements in understanding drought dynamics in a warming world.

1.8 Objective of the review

The primary objective of this review is to critically assess the influence of climate change on drought frequency, intensity, and duration across various geographical regions, with a particular focus on the application and comparative performance of standardised drought indices. By synthesising findings from global case studies, the review aims to highlight the

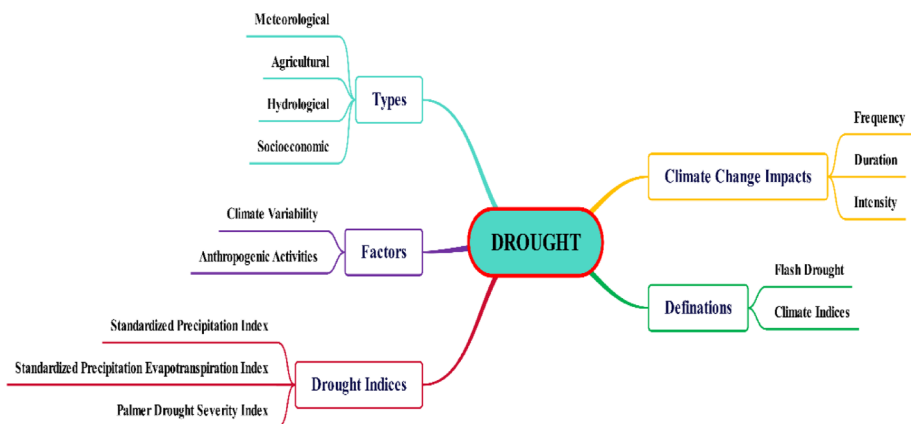


Fig. 2 Climate change and drought severity

evolving patterns of drought under changing climatic conditions and underscore the role of robust indices in monitoring, assessing, and predicting drought phenomena.

- *Evaluate the effectiveness of drought indices* Analyse various drought indices used to assess and monitor drought conditions, highlighting their advantages, limitations, and applicability in different climatic regions.
- *Examine the role of climate change in drought patterns* Investigate how climate change affects drought frequency, intensity, and duration across various geographical regions, utilising both historical and projected climate data.
- *Identify emerging trends in drought assessment and prediction* Explore advancements in drought forecasting models, remote sensing technologies, and machine learning approaches to enhance early warning systems and inform effective mitigation strategies.
- *Provide recommendations for future research and policy development* Suggest future directions for research on drought assessment, emphasising the need for improved methodologies, data integration, and policy frameworks to enhance climate resilience.

2 Methodology

2.1 Literature search and study selection

A systematic literature search was conducted across four major databases (Web of Science, Scopus, Google Scholar, and ScienceDirect) using keywords related to drought and climate change. The search encompassed publications from 2010 to 2024, with a particular emphasis on recent studies (2015–2024). The initial search yielded approximately 400 articles, which were filtered through title/abstract screening (200 articles) and full-text review to produce 169 final articles for analysis. Table 1 shows the complete search methodology and selection criteria. The inclusion criteria required peer-reviewed publications that focused on drought indices and climate change, with quantitative regional or global analyses. Quality assessment considered the journal's impact factor, citation count, and methodological rigour.

2.2 Regional classification and study distribution

Table 2 provides a quantitative summary of the geographical distribution of studies included in this review, indicating the number of research articles focused on specific regions and on a global/multi-regional scale.

2.3 Drought index selection and data sources

Table 3 summarises the various drought indices examined in this review, along with their commonly associated datasets and typical data sources. It highlights the reliance on satellite-derived and gridded climate data, particularly from sources such as CHIRPS, GPCC, ERA5, and MODIS, for calculating these indices across various studies.

Table 1 Systematic literature search methodology

Search component	Details
Databases searched	Web of Science, Scopus, Google Scholar, ScienceDirect
Search keywords/terms	“Drought indices”, “climate change”, “drought assessment”, “SPEI”, “SPI”, “PDSI”, “VCI”, “drought monitoring”, “meteorological drought”, “agricultural drought”, “hydrological drought”, “drought patterns”, “global warming”, “precipitation”, “evapotranspiration”
Time period covered	2010–2024 (primary focus on 2015–2024 for recent developments)
Geographic coverage	Global focus with studies from North America, Europe, Asia, Africa, South America, and Australia
Initial search results	~400 articles identified
After title/abstract screening	~200 articles
Final articles included	169 articles
Inclusion criteria	Published in peer-reviewed journals Focus on drought indices and climate change. Global or regional drought analysis
Quality assessment	Journal impact factor, citation count, and methodology rigour
Data extraction	Region studied, drought indices used, key findings, methodology, and time period

Table 2 Number of studies per region

Region	No. of studies
Amazon Basin	3
Australian Outback	2
Central Chile	1
Central Europe	15
Indian Subcontinent	10
Mediterranean	13
Northwest China	9
Southern Africa	7
Southwestern USA	6
Sub-Saharan Africa	9
Global/Multi-regional Studies	94

3 Drought indices

3.1 Overview of drought indices

Drought indices are crucial for assessing and tracking the severity of meteorological, agricultural, and hydrological droughts. The most effective indices for these drought types include rainfall deciles, soil moisture measurements, and total water deficit (Keyantash and Dracup 2002). Of the 111 drought indices reviewed, 44 are traditional, while 67 are based on remote sensing technology (Alahacoon and Edirisinghe 2022). Notably, meteorological drought

Table 3 Drought indices and their data sources

Drought index	Common dataset(s) used in studies	Typical data source
SPI	CHIRPS (Climate Hazards Group)— GPCC (Global Precipitation Climatology Centre)—TRMM, GPCP, IMD (India-specific), ERA5	Satellite + gridded precipitation datasets
SPEI	ERA5 (precipitation + temperature)— CRU TS4.06—TerraClimate	Reanalysis & gridded climate datasets
PDSI	CRU TS4.06 (precipitation + temperature)	Gridded historical climate data (TS series)
VCI	MODIS NDVI (MOD13Q1 / MOD13A2)	Vegetation Index from MODIS
TCI	MODIS LST (MOD11A2 / MOD11C3)	Land surface temperature from MODIS
VHI	Combined MODIS NDVI (MOD13) and LST (MOD11)	Derived using Kogan's method

monitoring relies on the highest proportion of traditional indices (20%), while agricultural drought monitoring uses the fewest (6.3%). However, 90% of remote sensing-based indices are applied to agricultural drought monitoring, highlighting the growing role of satellite technology in this field (Alahacoon and Edirisinghe 2022). The Standardised Precipitation Index (SPI) and the Standardised Precipitation-Evapotranspiration Index (SPEI) have demonstrated superior performance in capturing the impacts of drought on various hydrological, agricultural, and ecological variables (Vicente-Serrano et al. 2011). In conclusion, selecting a drought index depends on the type of drought being monitored and the available data. The Standardised Precipitation Index (SPI) and deciles/percentiles are generally considered the most appropriate for monitoring meteorological drought (Quiring 2009). The SPEI has demonstrated improved capability in identifying drought impacts compared to the SPI, especially during summer when drought-related impacts are most critical (Vicente-Serrano et al. 2012). As research advances, there is a growing trend to combine meteorological and remote sensing data for more thorough drought monitoring (Liu et al. 2016).

A wide range of drought indices has been developed to assess various types and aspects of drought. These include precipitation-based indices, such as SPI, climate-sensitive indices like SPEI, and satellite-based indices, including VCI and SMDI. Figure 3 provides an overview of commonly used drought indices, highlighting their key inputs and focus areas.

3.2 Classification of drought indices

Drought indices can be categorised differently, depending on their specific characteristics and applications. Traditional drought indices are typically location-specific or model-based, while remote sensing (RS) indices have gained prominence recently (Alahacoon and Edirisinghe 2022). Among traditional drought indices, meteorological drought monitoring utilises the highest number, with 22 indices, accounting for approximately 20% of the total, while agricultural drought monitoring has the fewest, with only 7 indices, or 6.3% (Alahacoon and Edirisinghe 2022). In contrast, 90% of remote sensing-based drought indices are used for agricultural drought monitoring, while only 10% are applied to hydrological and meteo-

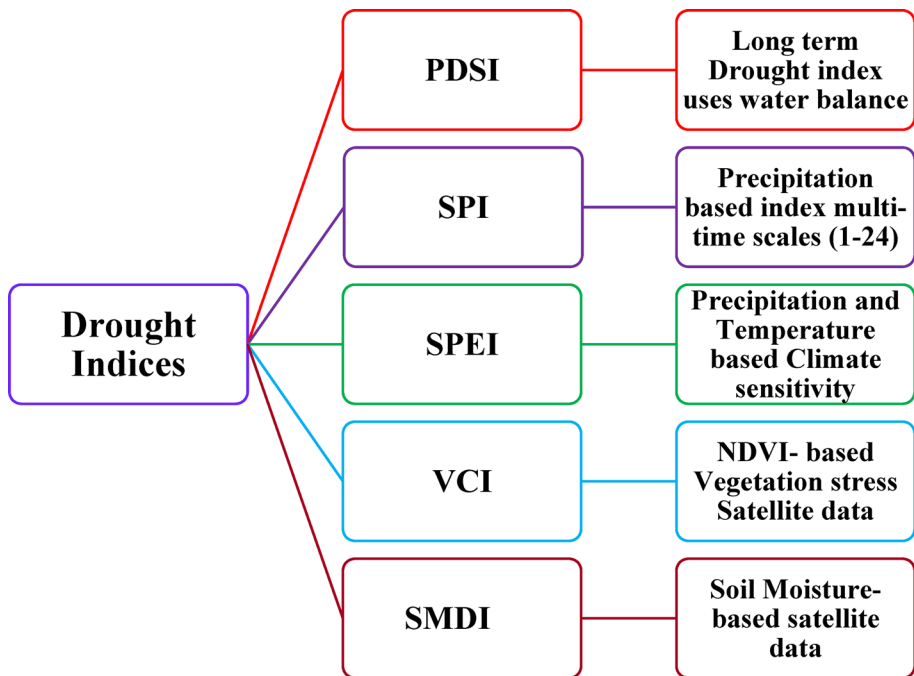


Fig. 3 Drought Indices Overview

rological droughts (Alahacoon and Edirisinghe 2022). This shift towards remote sensing (RS)-based indices for agricultural drought monitoring highlights the benefits of satellite technology, which offers data on spatial distribution and resolution. Drought indices can generally be categorised into four types: meteorological, agricultural, hydrological, and socio-economic (Vicente-Serrano et al. 2011). The selection of an index depends on the type of drought being monitored and the available data. While traditional indices remain essential, the growth of satellite technology has led to an increase in the development and application of remote sensing-based indices, especially for agricultural drought monitoring. This trend highlights the evolving nature of drought monitoring techniques and underscores the importance of utilising multiple indices for a comprehensive drought assessment.

3.2.1 Meteorological indices

Meteorological indices are crucial for quantifying and assessing various aspects of weather and climate, particularly drought and fire risk. Several studies have evaluated various indices for their effectiveness in characterising meteorological conditions and their associated impacts. For drought assessment, indices such as the Standardised Precipitation Index (SPI), the Standardised Precipitation Evapotranspiration Index (SPEI), rainfall deciles, and Z-scores are valuable (Keyantash and Dracup 2002; Shahabfar and Eitzinger 2013; Kim et al. 2017). These indices offer different strengths in capturing drought severity, frequency, and duration. Alternative probability distributions, such as log-normal or normal, can be used for SPI calculation at longer time scales (12 or 24 months), yielding results similar to those

of the gamma distribution. (Angelidis et al. 2012). The assumption of stationarity in the SPI calculation has been questioned due to the impacts of recent climate changes, leading to the development of a non-stationary version of the SPI. This version adjusts the probability distribution parameters and returns periods to account for changing climate conditions (Park et al. 2018). For instance, SPI and SPEI have been shown to overestimate drought severity and frequency, respectively, under certain conditions (Kim et al. 2017). The choice of the index may depend on the specific meteorological regime and the characteristics of the drought being studied. Interestingly, some studies have found that simpler indices, based solely on precipitation, can perform comparably to more complex indices that depict drought periods and intensity (Olukayode Oladipo 1985). It suggests that precipitation is a crucial factor in assessing meteorological drought. However, other research has demonstrated the value of incorporating multiple meteorological variables, such as temperature and evapotranspiration, particularly when studying drought duration and severity (Stagge et al. 2015b; Kim et al. 2017). The choice of potential evapotranspiration (PET) calculation method can impact SPEI results, with some methods exhibiting greater similarity to the standard FAO Penman–Monteith equation than others (Zarei and Mahmoudi 2020). In conclusion, selecting the appropriate meteorological indices depends on the specific application, regional climate, and availability of data. While some indices focus on single variables, such as precipitation, others incorporate multiple factors to assess meteorological conditions comprehensively. The ongoing development and evaluation of these indices contribute to an improved understanding and prediction of weather-related phenomena, including drought and fire risk.

3.2.2 Hydrological indices

Hydrological indices are crucial for assessing and monitoring water resources, particularly in the context of drought and climate change. Two such indices, the Streamflow Drought Index (SDI) and the Standardised Runoff Index (SRI), are used to analyse multi-step hydrological drought (Pathak and Dodamani 2016). These indices play a key role in quantifying and monitoring drought conditions, which can be complex and difficult to define. The PDSI employs a water balance model that accounts for previous conditions, which is considered an advantage over purely statistical indices (Vicente Serrano et al. 2011). A self-calibrating version, called the SC-PDSI, was developed to address inconsistent behaviour across different locations. This version adjusts the index's behaviour at any location by replacing empirical constants with dynamically calculated values (Wells et al. 2004). The Standardised Precipitation Index (SPI) and the Standardised Precipitation Evapotranspiration Index (SPEI) are also commonly used to assess the impact of climate variability on water availability (Lorenzo-Lacruz et al. 2010). While precipitation is a significant factor in explaining the temporal variability of hydrological parameters, the influence of temperature should not be overlooked. The SPEI, which incorporates precipitation and evapotranspiration, exhibits a slightly stronger response in river discharge and reservoir storage compared to the SPI (Lorenzo-Lacruz et al. 2010). It underscores the importance of considering multiple factors when evaluating hydrological conditions. In conclusion, hydrological indices provide valuable insights for water resource management and policy planning. They can be used to evaluate the performance of water systems under various climate change scenarios (Marques et al. 2022) and to assess the sustainability of river basins (Sediqi and Komori 2024). However, it is crucial to consider the limitations and uncertainties associated with

these indices, especially when using them for risk-based decision-making in water resource systems (McInerney et al. 2024).

3.2.3 Agricultural indices

Agricultural drought indices are vital tools for monitoring and assessing the impact of drought on crop production. Several indices have been developed and evaluated for their ability to capture different aspects of agricultural drought. The Standardised Precipitation Evapotranspiration Index (SPEI) is considered one of the most effective indices for representing soil moisture conditions and crop yields (Tian et al. 2018). It strongly correlates with crop yields alongside Palmer's Z-index (Tian et al. 2018). However, the performance of these indices can vary depending on the crop type and growth stage, suggesting that no single index can fully capture all dimensions of agricultural drought (Tian et al. 2018). Notably, soil moisture-based indices, such as the Soil Water Deficit Index (SWDI) and the Soil Moisture Agricultural Drought Index (SMADI), have shown higher correlations with widely recognised agricultural drought indices, including the Atmospheric Water Deficit (AWD) and the Crop Moisture Index (CMI) (Pablos et al. 2017). Combining multiple indices, such as the Integrated Drought Condition Index (IDCI), which merges SPEI, the Vegetation Cover Index (VCI), and the Soil Moisture Condition Index (SMCI), has proven to improve agricultural drought monitoring (Li et al. 2024). In conclusion, while various agricultural drought indices exist, their effectiveness depends on crop type, growth stage, and regional characteristics. Integrating multiple indices and using remote sensing data are emerging trends in agricultural drought monitoring (Liu et al. 2016). Choosing the right index should consider the specific context and needs of the agricultural system being studied. The Modified Palmer Drought Severity Index (MPDSI) has been proposed to incorporate agricultural irrigation data, improving drought monitoring in irrigated regions (Yu et al. 2019). Different soil moisture-based indices may have varying sensitivities and applicability, depending on the region and specific drought conditions being monitored, with some showing lower correlation with established drought indices, such as the atmospheric water deficit (AWD) and crop moisture index (CMI) (Pablos et al. 2017).

3.2.4 Remote sensing-based drought indices

Remote sensing-based drought indices have become crucial for monitoring and assessing drought conditions across various ecosystems. Indices such as the Vegetation Condition Index (VCI), Temperature Condition Index (TCI), and Vegetation Health Index (VHI) are commonly derived from satellite data to detect meteorological and agricultural droughts (Zou et al. 2020; Lima et al. 2023). These indices utilise data from sensors like MODIS to analyse vegetation health, land surface temperature, and precipitation patterns. (Rhee et al. 2010; Li et al. 2020). Interestingly, the performance of these indices varies depending on the ecosystem and timing. For instance, TCI outperforms VCI and VHI in monitoring seasonal and annual droughts in tropical dry forests (Zou et al. 2020). However, in both arid and humid regions, combined indices such as the Scaled Drought Condition Index (SDCI) have demonstrated better performance than single indices (Rhee et al. 2010; Li et al. 2020). Under global greening trends, VCI may reflect changes in NDVI, which could compromise its ability to assess long-term drought trends accurately (Yin et al. 2024). The

relationship between VCI and meteorological drought indices can vary across regions, influenced by climate, land use/land cover, irrigation practices, and soil properties (Quiring and Ganesh 2010). The VHI has demonstrated an optimal correlation with both short-term and long-term precipitation in continental China (Li et al. 2020). Remote sensing-based drought indices have proven to be valuable tools for monitoring and assessing drought conditions in Southern America and Africa. These indices provide comprehensive, continent-wide assessments with high spatial and temporal resolutions, which are particularly useful in regions with limited in-situ data (Winkler et al. 2017). In Africa, the Standardised Precipitation Index (SPI) derived from TRMM data and the Vegetation Condition Index (VCI) based on MODIS data have been successfully utilised to analyse the spatiotemporal patterns of droughts. These indices revealed major drought years in eastern Africa (2009 and 2011) and southern Africa (2003 and 2015/2016). The study also observed a connection between El Niño events and drought episodes in southern Africa, although no universal ENSO-related drought pattern could be established (Winkler et al. 2017). Interestingly, different indices show varying sensitivities to drought conditions. For instance, the Temperature Condition Index (TCI) was found to be more sensitive during dry seasons or months with high temperatures, while the VCI detected drought more sensitively during rainy seasons (Ezzahra et al. 2023). The Evaporative Stress Index (ESI) has demonstrated higher correlations with anomalies in maize and wheat yields compared to other indices, indicating its potential for monitoring crop impacts (Ghazaryan et al. 2020). These indices provide valuable insights into drought patterns, their relationship with climate phenomena such as ENSO, and their impact on agriculture. The integration of multiple indices and their combination with phenometrics holds promise for enhancing drought monitoring and early warning systems in these regions (Winkler et al. 2017; Ghazaryan et al. 2020). In conclusion, remote sensing-based drought indices offer significant benefits for real-time drought monitoring and early warning systems, particularly in regions with limited data availability (Ejaz et al. 2023). While each index has its strengths and limitations, combining multiple indices in integrated approaches often yields more comprehensive drought assessments across different climate regions (Jiao et al. 2019). The development of indices such as the Geographically Independent Integrated Drought Index (GIIDI) and the Comprehensive Agriculture Drought Condition Indicator (CADCI) holds promise for enhancing drought monitoring capabilities across diverse biomes and climates (Jiao et al. 2019; Alkaraki and Hazaymeh 2023).

4 Assessment of climate change impact on drought using indices

Drought assessment in the context of climate change relies on multiple indices, including hydrological, meteorological, agricultural, and remote sensing-based approaches. Each of these captures different dimensions of drought characteristics and drivers. Figure 4 summarises the key types of indices used to evaluate the impacts of climate change on drought, highlighting their inputs and observational sources.

4.1 Studies using meteorological indices

Climate change has a significant impact on drought conditions globally, underscoring the need for various meteorological drought indices to ensure accurate assessments. The Stan-

standardised Precipitation Index (SPI) and the Standardised Precipitation Evapotranspiration Index (SPEI) are widely used to quantify rainfall deficits and evaluate drought characteristics (Diani et al. 2019; Stagge et al. 2015a, b). These indices can be applied at multiple timescales to assess different drought types, with agricultural impacts typically explained by 2–12 month anomalies and energy-related impacts by 6–12 month anomalies (Stagge et al. 2015a, b). However, research has shown that relying on a single index may not adequately monitor drought evolution in the context of climate change. For example, the SPEI, which accounts for precipitation and temperature, may be more effective than SPI alone in warming scenarios (Marcos-Garcia et al. 2017). Additionally, the effects of climate change on drought intensity, frequency, and duration tend to escalate from meteorological drought to agricultural drought to hydrological drought (Wang et al. 2015). The Amazon River Basin (ARB) plays a crucial role in the global hydrological cycle, and understanding drought patterns in this region is of vital importance. Several drought indices have been employed to assess drought conditions in the Amazon region. The Standardised Precipitation-Evapotranspiration Index (SPEI) at multiple time scales (3, 6, and 12 months) has been utilised to characterise dry spells and extreme drought events in the ARB from 1901 to 2018 (Pardo-Loaiza et al. 2021). This study revealed a weak basin-wide drying trend, although no significant trend in extreme drought events was observed in terms of spatial coverage, intensity, and duration. However, a transition to drier-than-normal conditions has been evident since the 1970s, coinciding with changes in climate oscillation patterns (Pardo-Loaiza et al. 2021). Interestingly, when comparing various drought indices in the Amazon Basin, the Standardised Runoff Index (SRI) has been used as a benchmark for meteorological drought indices, such as scPDSI, SPI, and SPEI (Joetzjer et al. 2013). The study found that the simple SPI, based solely on precipitation, was no less suitable than more sophisticated meteorological drought indices for detecting interannual SRI variations in the Amazon Basin (Joetzjer et al. 2013). While various drought indices have been applied to the Amazon region, there is no clear consensus on a single best index. Assess drought conditions in this critical ecosystem. In conclusion, assessing the impacts of climate change on drought necessitates a multifaceted approach, utilising multiple indices and considering various time scales. Drought indices should be developed to address the non-stationarity of climate variables and human influences (Han and Singh 2023). Moreover, incorporating global circulation indices and teleconnection patterns can enhance understanding regional

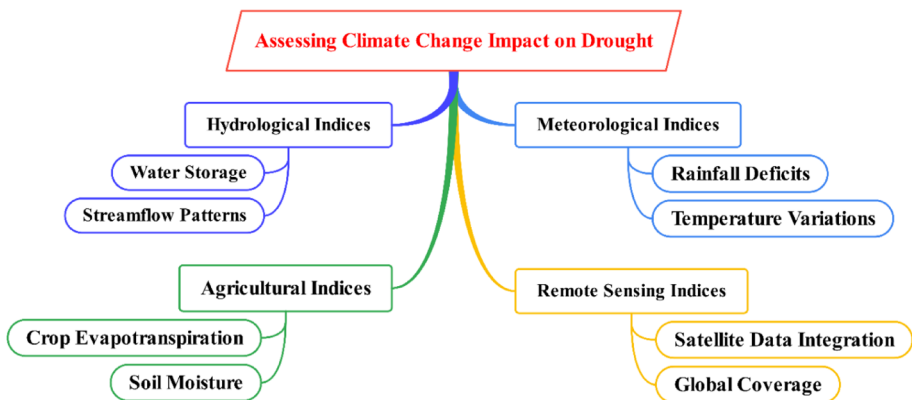


Fig. 4 Assessing climate change's impact on drought

drought dynamics (Ogunrinde et al. 2024). As climate change continues to alter precipitation patterns and raise temperatures, it is essential to adopt robust drought assessment methods for effective water resource management and climate adaptation strategies.

4.2 Studies using hydrological indices

Climate change is expected to significantly impact drought conditions, with studies indicating that more severe and frequent droughts will occur in many regions. Multiple papers have investigated this using hydrological drought indices such as the Standardised Streamflow Index (SSI) and the Streamflow Drought Index (SDI). (Feyen and Dankers 2009; Chattopadhyay et al. 2017; Giri et al. 2021). Research indicates that hydrological droughts are expected to become more severe and persistent in most parts of Europe, except the northernmost regions, by the end of this century (Feyen and Dankers 2009). Similarly, projections for the Kentucky River basin suggest that increasing the maximum duration of hydrological droughts may be necessary, although drought intensity may decrease under future climate scenarios (Chattopadhyay et al. 2017). Significant increases in drought magnitude and severity are expected in South Korea across different time scales (Nam et al. 2015). Interestingly, the impact of climate change on drought appears to vary depending on the type of drought being assessed. For example, one study found that the effects of climate change become more pronounced, progressing from meteorological to agricultural to hydrological drought, with agricultural drought being particularly sensitive to climate change (Wang et al. 2019). In another study conducted in a semiarid watershed in northern China, projections suggested less frequent but more severe hydrological droughts at various warming levels (1.5 °C, 2 °C, and 3 °C) (Jiao and Yuan 2019). In conclusion, while there is general agreement that climate change will exacerbate drought conditions, the specific impacts may vary regionally and depend on the type of drought assessed. These findings underscore the importance of using multiple drought indices and considering various climate scenarios when assessing future drought risks for effective water resource management and adaptation strategies.

4.3 Studies using agricultural indices

Agricultural drought indices are becoming increasingly important for evaluating the effects of climate change on drought conditions, providing valuable insights into the complex relationships between climate, soil moisture, and crop productivity (Tigkas et al. 2015; Wang et al. 2015). These indices provide a more comprehensive understanding of how drought affects agriculture, in contrast to traditional meteorological drought indices. One such example is the Crop Reconnaissance Drought Index (CRDI), a modified version of the widely used Reconnaissance Drought Index (RDI). The CRDI replaces reference evapotranspiration with crop-specific evapotranspiration, enabling a more targeted assessment of drought impacts on specific crops, particularly in Mediterranean regions (Tigkas et al. 2020). Interestingly, research has shown that agricultural drought indices, such as the Standardised Soil Water Index (SSWI), are more sensitive to climate change than meteorological or hydrological drought indices (Wang et al. 2011). This heightened sensitivity is attributed to the complex interactions between temperature increases, precipitation changes, and nonlinear hydrological responses during the crop growing season. Recent research has

focused on linking meteorological drought indices with on-the-ground impacts, including agricultural effects (Stagge et al. 2015a, b). Studies have shown that agricultural impacts are often linked to anomalies lasting 2–12 months, with longer anomalies generally associated with specific agricultural management practices. The development of drought impact models with strong predictive capabilities highlights the potential of using these models to forecast the likelihood of drought impacts based on meteorological indices. In conclusion, agricultural drought indices are crucial tools for evaluating the impact of climate change on drought conditions and crop productivity. By incorporating crop-specific factors, such as evapotranspiration and soil moisture, these indices provide a more detailed understanding of drought's impact on agriculture. As climate change continues to influence precipitation patterns and raise temperatures, these indices will become increasingly essential for monitoring, forecasting, and adapting to drought in the agricultural sector.

4.4 Studies using remote sensing-based indices

Remote sensing-based drought indices have become increasingly important in assessing the impact of climate change on drought patterns. These indices provide continuous global coverage and offer valuable insights into various hydro-meteorological variables that influence drought conditions (Khorrami and Gündüz 2022). The use of remote sensing data for drought monitoring has grown significantly, with 67 out of 111 reviewed drought indices based on remote sensing techniques (Alahacoon and Edirisinghe 2022). This shift towards satellite-based indices is particularly evident in agricultural drought monitoring, where 90% of the indices utilise remote sensing data (Alahacoon and Edirisinghe 2022). The availability of new remote sensing data has shifted the focus of drought assessments from crop or field scale to large-scale evaluations (Natale et al. 2023).

Several studies have highlighted the effectiveness of remote sensing-based drought indices. For example, the GRACE-based Water Storage Deficit Index (WSDI) demonstrated strong temporal and spatial correlations with the traditional Standardised Precipitation Index (SPI) over Turkey (Khorrami and Gündüz 2022). Similarly, the Evapotranspiration Condition Index (ETCI), derived from actual evapotranspiration data, showed strong performance in drought monitoring, with a correlation of 0.655 with the three-month SPI (Behifar et al. 2023). However, challenges remain in developing robust drought indices for reliable assessment under changing climate conditions. Factors such as selecting an appropriate baseline period, accurately estimating potential evapotranspiration, and combining multiple drought indices must be considered (Han and Singh 2023). Additionally, quantifying anthropogenic impacts on drought assessment is crucial for adapting to changing drought conditions and minimising human-induced drought (Han and Singh 2023). In conclusion, remote sensing-based drought indices have proven to be valuable tools for assessing the impact of climate change on drought patterns. The ability of remote sensing-based drought indices to offer continuous global coverage and integrate multiple hydro-meteorological variables makes them especially valuable for large-scale drought monitoring and assessment. However, further research is necessary to overcome challenges in developing reliable drought indices amid a changing climate and to deepen our understanding of the complex interactions between climate change and drought patterns.

5 Strengths and limitations of different indices

5.1 Comparison of index performance in different climatic regions

Drought indices vary in effectiveness across different climatic regions and in response to climate change. For instance, in Iran, the Z-score, China-Z Index (CZI), and Modified CZI (MCZI) were identified as effective predictors of meteorological drought, offering alternatives to the Standardised Precipitation Index (SPI) due to data limitations (Shahabfar and Eitzinger 2013). In a semi-arid basin in western India, the Standardised Precipitation Evapotranspiration Index (SPEI) at the 9-month scale (SPEI-9) was the most accurate, followed by the Reconnaissance Drought Index (RDI-9) and SPI-9 (Wable et al. 2019). In European forests, drought indices showed varying performance across bioclimatic zones, with temperate and Mediterranean forests more responsive to short- and intermediate-term droughts (<1 year). In contrast, continental forests were more sensitive to long-term droughts (Bhuyan et al. 2017). In the Continental United States (CONUS), meteorological drought indices generally indicated more severe drought conditions than those indicated by agricultural indices. The Soil Moisture Condition Index (SMCI) was strongly correlated with short-term SPI but was less effective in heavily forested areas. Meanwhile, the Precipitation Condition Index (PCI) demonstrated stronger correlations with short-term drought conditions across most climate divisions (Zhang et al. 2016).

A comparative evaluation of drought indices is crucial for selecting appropriate tools tailored to specific types of droughts. Figure 5 presents a comprehensive comparison of drought index performance across ten global regions, showing accuracy, sensitivity, specificity, and correlation metrics for seven widely-used drought indices (CDI, PDSI, SPEI, SPERI, SPI, TCI, and VCI). The heatmap visualisation reveals distinct regional patterns in index effectiveness, with notable variations in performance across different climatic zones.

Figure 6 synthesises the average performance scores across all evaluation metrics for each drought index by region. This consolidated view highlights that SPEI consistently performs well across most regions (average scores 0.80–0.89). At the same time, PDSI exhibits more variable performance, particularly in tropical regions such as the Amazon Basin (0.72) and Southern Africa (0.69).

5.2 Sensitivity to climate change

Climate change affects the performance of drought indices, particularly those that assume stationarity in climatic variables. In the Mediterranean Basin, the SPI was found to potentially underestimate future meteorological drought severity because it does not account for temperature changes. At the same time, SPEI and SPI showed increased drought intensity and magnitude under climate change scenarios (Marcos-García et al. 2017). The sensitivity of drought indices to precipitation (P) and reference evapotranspiration (ET_o) varies. The Palmer Drought Severity Index (PDSI) showed the lowest sensitivity to climate inputs. At the same time, the SPEI demonstrated the highest sensitivity to ET_o variation, with clear geographic patterns influenced by aridity (Vicente-Serrano et al. 2014). It suggests that SPEI may be more suitable for detecting the impacts of climate change on drought severity across diverse climatic conditions. SPEI offered the best representation of drought patterns in global arid zones among indices calculated from CMIP6 historical simulations (Wu et

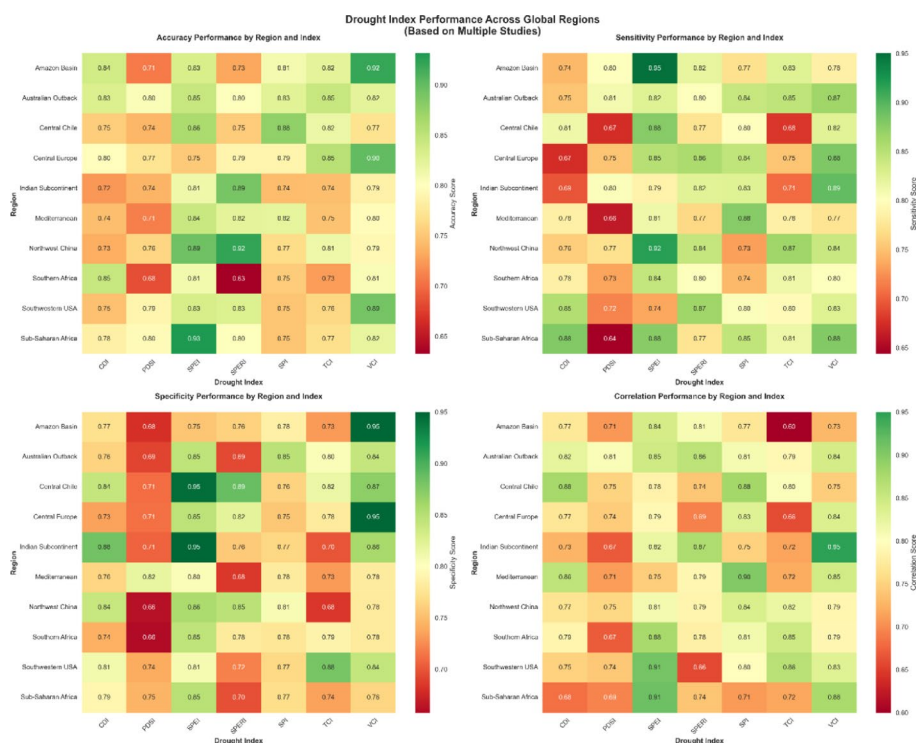


Fig. 5 Comparative performance of seven drought indices CDI, PDSI, SPEI, SPERI, SPI, TCI, and VCI across ten global regions based on four evaluation metrics: accuracy (top left), sensitivity (top right), specificity (bottom left), and correlation (bottom right)

al. 2024). However, formulating drought indices without considering background state factors may lead to drought artefacts under a warming climate. Energy budget-based potential evapotranspiration (PET) estimation is recommended over temperature-based equations to avoid overestimating future drying (Mukherjee et al. 2018). In conclusion, the performance and sensitivity of drought indices vary across climatic regions and in response to climate change. Careful selection of indices that are appropriate for specific regional characteristics and climate change considerations is crucial for accurate drought assessment and management.

5.3 Data requirements and availability

Drought indices vary in their strengths and limitations, particularly regarding data requirements, availability, and applicability to different types of droughts. The availability and quality of data heavily influence the effectiveness of these indices. For example, the Standardised Precipitation Index (SPI) and deciles/percentiles are suitable for monitoring meteorological drought because they are simple to use and require minimal data. (Quiring 2009). In contrast, more complex indices, such as the Palmer Drought Severity Index (PDSI), require multiple input parameters, which can be challenging to obtain in regions with limited data. (Yang et al. 2017). Remote sensing-based indices have emerged as useful tools to

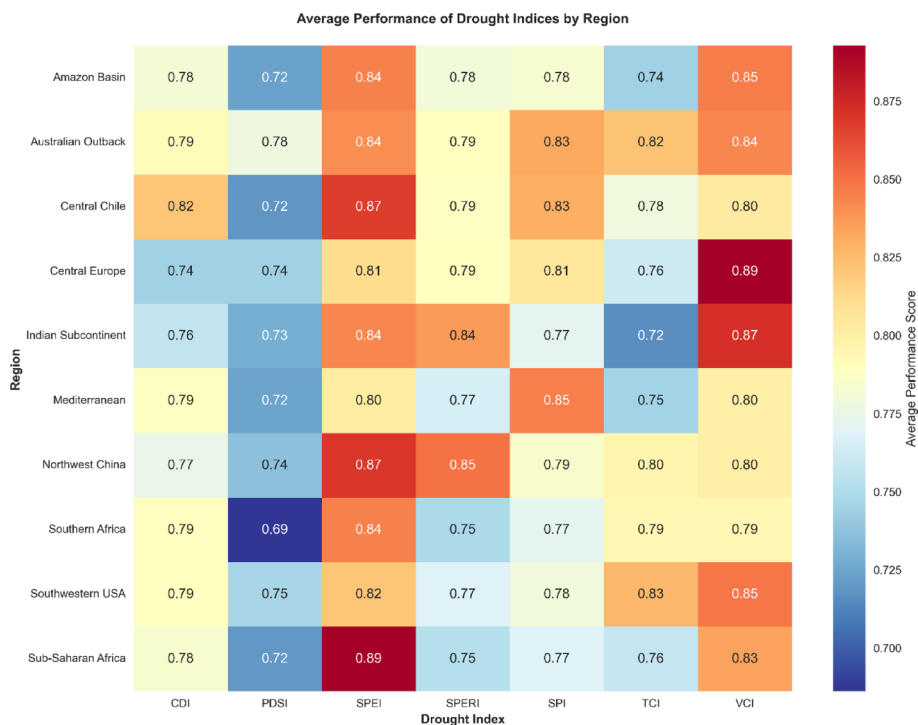


Fig. 6 Heatmap showing the average performance scores of seven widely used drought indices, CDI, PDSI, SPEI, SPERI, SPI, TCI, and VCI, across ten global regions

address the challenge of limited ground-based data, enabling the calculation of meteorological, agricultural, and hydrological drought indices (Sseguya and Jun 2024). Different indices are also suited to various types of droughts. For meteorological drought, the Reconnaissance Drought Index (RDI) has shown promise (Sseguya and Jun 2024). Agricultural drought can be effectively monitored using indices such as the Soil Moisture Agricultural Drought Index (SMADI) and the Vegetation Condition Index (VCI) (Wei et al. 2021). The Streamflow Drought Index (SDI) is particularly effective for assessing hydrological drought (Sseguya and Jun 2024). Figure 7 contrasts meteorological and agricultural indices, highlighting their data requirements, climate sensitivity, and suitability for drought monitoring applications.

Interestingly, the Standardised Precipitation-Evapotranspiration Index (SPEI) calculated over the prior 12 months is the most predictive for simulating long-term soil moisture patterns, which is crucial for understanding ecological impacts in dryland regions (Barnard et al. 2021). In conclusion, selecting the appropriate drought index depends on the specific type of drought being monitored, the availability of data, and the regional characteristics of the area in question. Combined drought indices, such as the Multi-Index Drought (MID) model, have shown superior performance by integrating the strengths of various individual indices (Sun et al. 2012). However, the complexity of drought necessitates ongoing research to develop more comprehensive and adaptable indices, especially in the context of climate change (Yihdego et al. 2019). Table 4 provides a synthesised overview of key drought indi-

ces applied regionally, along with significant findings from recent studies conducted across diverse geographical settings.

6 Emerging trends and future directions

In recent years, novel drought indices have been developed to improve monitoring precision and better account for climate variability and ecosystem responses. Figure 8 highlights the emergence of new drought indices that integrate advanced satellite data, machine learning algorithms, and region-specific variables for enhanced prediction and early warning.

6.1 Development of new indices and integration of multiple data sources

Developing new drought indices and integrating multiple data sources are key trends in improving drought monitoring and assessment, addressing the limitations of existing methods and enhancing accuracy across different regions. New indices are designed to overcome geographical constraints and improve performance in various climate zones. For example, the Geographically Independent Integrated Drought Index (GIIDI_station) utilises geographically weighted regression and principal component analysis to combine various vegetation and meteorological indices from multiple remote sensing platforms and ground-based data (Jiao and Yuan 2019). This method shows better correlations with in-situ drought indices and performs effectively across diverse environmental conditions. Similarly, the Linear Combined Index (LCI) integrates multiple standardised indices, such as the Standardised Precipitation Index (SPI), Standardised Soil Moisture Index (SSI), and Standardised Runoff Index (SRI), offering a more comprehensive approach to drought monitoring (Hao et al. 2016). Another major trend is the integration of various data sources, combining remote sensing with ground-based observations to create high-resolution drought index datasets. For example, a study in Spain developed a dataset of seven drought indices with a spatial resolution of 1.1 km and weekly temporal resolution, incorporating multiple data sources (Vicente-Serrano et al. 2017). In another study, machine learning techniques were employed to merge multi-source remote sensing data from MODIS, GPM, and GLDAS to recreate the Standard Precipitation Evapotranspiration Index (SPEI), highlighting the potential of data-driven models in drought monitoring (Zhao et al. 2022b). In conclusion, the future of

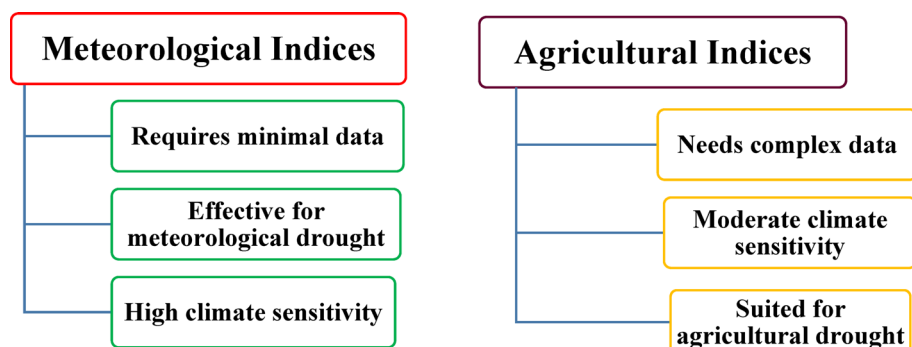


Fig. 7 Comparing drought indices for effective monitoring

Table 4 Regional drought indices and key findings

Region	Drought indices used	Key findings
Africa	Standardized Precipitation Index (SPI) Standardized Precipitation Evapotranspiration Index (SPEI) Vegetation Condition Index (VCI) Combined Drought Indicator (CDI) Global Precipitation Climatology Centre-Drought Index (GPCC-DI)	Drought frequency and intensity are increasing across major river basins in Southern Africa at specific global warming levels (Abiodun et al. 2019) Multi-scale SPEI analysis (1960–2018) reveals spatiotemporal drought patterns with significant agricultural impacts (Ogunrinde et al. 2025) ENSO teleconnections significantly influence drought patterns, with rainfall anomalies and vegetation conditions closely linked (Winkler et al. 2017) Small-scale farmers face increasing vulnerability to drought impacts, affecting food security and livelihoods (Lottering et al. 2021)
Europe	Standardized Precipitation Index (SPI) Standardized Precipitation Evapotranspiration Index (SPEI) Palmer Drought Severity Index (PDSI) Combined Drought Indicator (CDI) Reconnaissance Drought Index (RDI)	Increasing drought severity caused by temperature rise in southern Europe, with environmental drought risk escalating under global warming (Vicente-Serrano et al. 2014) Mediterranean drought frequency has increased significantly, requiring re-thinking of water policy priorities (Hoerling et al. 2012; Iglesias et al. 2011) Mid-latitude European stations show pronounced trends in drought occurrence and severity (1951–2015) using SPI and SPEI indices (Dukat et al. 2022) Diverging hydrological drought traits observed with global warming, indicating changing drought characteristics (Cammalleri et al. 2020)
Asia (East/Southeast)	Standardized Precipitation Index (SPI) Standardized Precipitation Evapotranspiration Index (SPEI) Vegetation Condition Index (VCI) Temperature Condition Index (TCI) Global Precipitation Climatology Centre-Drought Index (GPCC-DI)	More severe hydrological drought events emerge at different warming levels, particularly in northern China watersheds (Jiao and Yuan 2019) Climate change impacts on meteorological, agricultural and hydrological droughts show varying intensities across China (Leng et al. 2015) Monsoon drought variability over India shows significant temporal and spatial patterns affecting agricultural productivity (Niranjan Kumar et al. 2013) Combined statistical and spatially distributed models prove effective for evaluating future drought conditions (Kang and Sridhar 2017)
North America	Standardised Precipitation Index (SPI) Palmer Drought Severity Index (PDSI) Standardised Precipitation Evapotranspiration Index (SPEI) Vegetation Condition Index (VCI) Self-Calibrating Palmer Drought Severity Index (scPDSI)	Spatial and temporal drought patterns across the Continental US show significant variations during the past century with increasing anthropogenic influences (Ge et al. 2016) Multiple remote sensing-based drought indices effectively monitor agricultural drought, with VCI showing strong correlation with meteorological conditions (Rhee et al. 2010) Great Plains low-level jet changes relate to recent drought patterns and associated precipitation changes (Barandiaran et al. 2013) Urban adaptation strategies for mega-drought require anticipatory water modeling and policy planning (Gober et al. 2016)

Table 4 (continued)

Region	Drought indices used	Key findings
South America	Standardized Precipitation Index (SPI) Standardized Precipitation Evapotranspiration Index (SPEI) Vegetation Condition Index (VCI) Enhanced Vegetation Index (EVI) Palmer Drought Severity Index (PDSI)	Amazon rainforest shows pronounced loss of resilience since the early 2000s, with drought sensitivity increasing significantly (Boulton et al. 2022) Statistically downscaled CMIP6 projections indicate changing precipitation and hydrological drought patterns across South America (Ferreira et al. 2023) Mega-drought in Central Chile demonstrates local perception impacts and adaptation challenges under changing climate conditions (Aldunce et al. 2017) Fish mortality during the 2023 Amazonian drought highlights ecosystem vulnerability and climate change impacts (Braz-Mota and Luis Val 2024)
Middle East	Standardized Precipitation Index (SPI) Standardized Precipitation Evapotranspiration Index (SPEI) Vegetation Condition Index (VCI) Temperature Condition Index (TCI) Combined Agriculture Drought Condition Indicator (CADCI)	Comprehensive remote sensing-based Agriculture Drought Condition Indicator using machine learning shows effectiveness in the Kingdom of Saudi Arabia (Alkaraki and Hazaymeh 2023) Spatio-temporal analysis using Global Precipitation Climatology Centre-Drought Index reveals dry and wet period patterns in Iran (Hosseini et al. 2021) High Atlas regions show effective variance-based fusion of VCI and TCI for agriculture drought classification (Ezzahra et al. 2023) Long-term CHIRPS precipitation analysis enables effective drought monitoring over Southeastern Iran (Mianabadi et al. 2022)
Australia/Oceania	Standardized Precipitation Index (SPI) Soil Moisture Drought Index Standardized Precipitation Evapotranspiration Index (SPEI) Vegetation Condition Index (VCI) Deciles Index	High spatial resolution soil moisture data proves effective for drought monitoring across Australia (2015–2019) with improved temporal and spatial coverage Multiple remote sensing drought indices show varying effectiveness for monitoring drought in the Greater Changbai Mountains region Remote sensing and traditional drought monitoring indices assessment reveal global and regional scale applicability differences (Alahacoon and Ediris-inghe 2022)
Global/multi-regional	Standardized Precipitation Index (SPI) Standardized Precipitation Evapotranspiration Index (SPEI) Palmer Drought Severity Index (PDSI) Vegetation Condition Index (VCI) Combined Drought Indicator (CDI)	Evidence of anthropogenic impacts on global drought frequency, duration, and intensity with significant increasing trends (Chiang et al. 2021) Global warming and drought relationship shows complex patterns with regional variations and increased risk under climate change scenarios World drought frequency, duration, and severity analysis (1951–2010) reveals increasing global drought trends with regional hotspots (Spinoni et al. 2014) Observation-constrained projections reveal longer-than-expected dry spells globally, indicating intensifying drought conditions (Petrova et al. 2024)

drought monitoring lies in developing integrated, geographically independent indices and the fusion of multiple data sources using advanced techniques such as machine learning. These approaches aim to provide more accurate, comprehensive, and adaptable drought assessments across diverse regions and changing climate conditions.

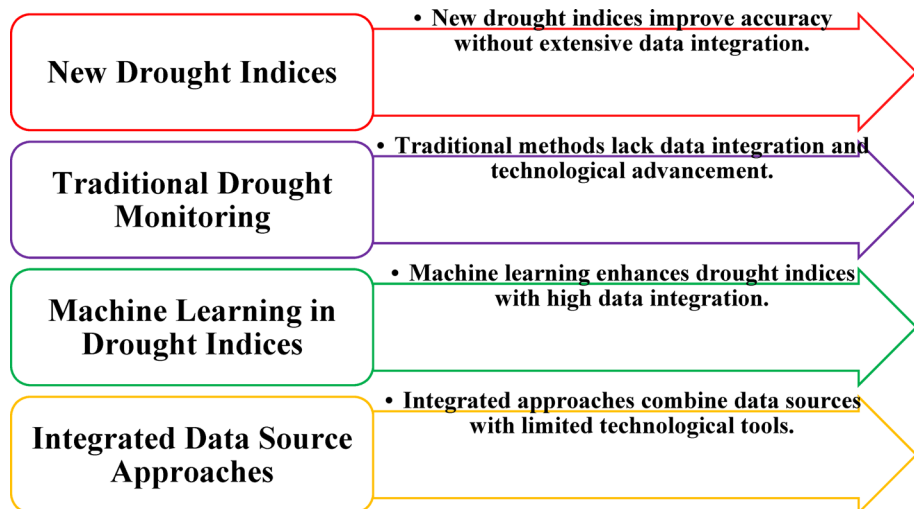


Fig. 8 Advancements in drought monitoring and prediction

6.2 Incorporation of machine learning and AI techniques

The integration of machine learning (ML) and artificial intelligence (AI) into drought assessment is an emerging trend in climate science, providing advanced tools for drought monitoring, classification, and forecasting (Bagwan and Gavali 2021; Prodhon et al. 2022). Over the past decade, ML/AI techniques have increasingly enhanced traditional drought indices like SPI, SPEI, and PDSI, significantly improving model robustness and predictive accuracy (Prodhon et al. 2022). ML/AI models outperform traditional statistical approaches in handling multivariate datasets, including climatic, hydrological, and remote sensing variables. For instance, SVM achieved the lowest RMSE (0.031), followed by RF (0.034). In contrast, GPR models yielded high R^2 values of 0.95 and 0.93 for SPI-3 and SPI-6 predictions, respectively (Prodhon et al. 2022; Hassan and Wang 2024). Deep learning models also reported ROC AUC scores of 0.948 for one-month forecasts, marking a 54% improvement over classical methods (Osmanlı et al. 2024). Supervised algorithms, such as RF, SVM, GBM, and LSTM networks, effectively capture nonlinear patterns in drought data. LSTM has outperformed ARIMA for long-term SPI forecasting, especially at 6- and 12-month scales (Elbeltagi and Massoud 2023; Osmanlı et al. 2024), with LSTM-CM improving accuracy from 29.17 to 54.29% for 1-month lead times (Park et al. 2023). Hybrid models integrating ML with remote sensing (e.g., NDVI, SMAP) enable real-time drought detection and high-resolution monitoring, surpassing traditional single-variable models (Zhang and Wang 2025). ML is also used to classify drought intensity and map drought risks, with RF and SVM achieving classification accuracies of over 90% (Ahmad and Liu 2024; Hassan and Wang 2024). Nonetheless, challenges persist, including model transparency (Karbasi et al. 2023), data scarcity in certain regions, and the high computational demands of deep learning (Zhang and Wang 2025). Standardisation and validation across regions remain crucial for the broader deployment of AI-driven drought early warning systems. In summary, ML and AI offer measurable advancements in drought modelling, improving accuracy by

15–50%, extending forecast lead times by 1–2 months, and enhancing spatial resolution. Future research should focus on developing interpretable, scalable models and fostering collaboration between AI experts and climate scientists.

6.3 Improved spatial and temporal resolution

Recent studies have focused on developing drought indices with improved spatial and temporal resolution to enhance drought monitoring capabilities. Several researchers have created high-resolution datasets and monitoring systems to address this need. In China, a high-resolution daily SPI/SPEI raster dataset was developed covering the period from 1979 to 2018, with a spatial resolution of $0.1^\circ \times 0.1^\circ$ and a temporal resolution of one day (Zhang et al. 2022).

This dataset demonstrated high accuracy and credibility, providing improved results compared to monthly SPI/SPEI. Similarly, a high-resolution drought index dataset in Spain was created from 1961 to 2014, with a spatial resolution of 1.1 km and a weekly temporal resolution (Vicente-Serrano et al. 2017). Spain's real-time drought monitoring system was also established, providing quasi-weekly updates with a spatial resolution of 1.1×1.1 km (Vicente-Serrano et al. 2023). Notably, some studies have focused on integrating multiple indices to enhance the accuracy of drought monitoring. For example, a methodology combining the Standardised Precipitation-Evapotranspiration Index and Soil Moisture Anomalies has been proposed for monitoring agricultural drought at the municipal level in Italy (Rossi et al. 2023). Another study developed a multi-index drought (MID) model for assessing agricultural drought risk on the Canadian Prairies, which automatically selects and combines optimal drought indices (Sun et al. 2012). In conclusion, these advancements in the spatial and temporal resolution of drought indices, as well as the integration of multiple indices, have significantly improved drought monitoring capabilities. These high-resolution datasets and monitoring systems provide more accurate and timely information for drought management and mitigation strategies across various regions.

7 Challenges and research gaps

Despite progress in drought assessment and monitoring, several challenges remain that hinder effective policy formulation and climate adaptation planning. These include uncertainties in climate projections, lack of standardisation in index selection, scale issues, and insufficient long-term observations. Figure 9 illustrates the key challenges and research gaps that need to be addressed to enhance the reliability and applicability of drought studies globally.

7.1 Uncertainty in climate projections

Climate projections indicate an increase in the frequency and intensity of droughts in many regions worldwide; however, significant uncertainties remain in these projections (Petrova et al. 2024). This uncertainty limits the accuracy of drought forecasts and regional projections (Barriopedro et al. 2023). A significant challenge is the incomplete understanding of the dynamic processes, regional drivers, and feedback mechanisms that influence drought

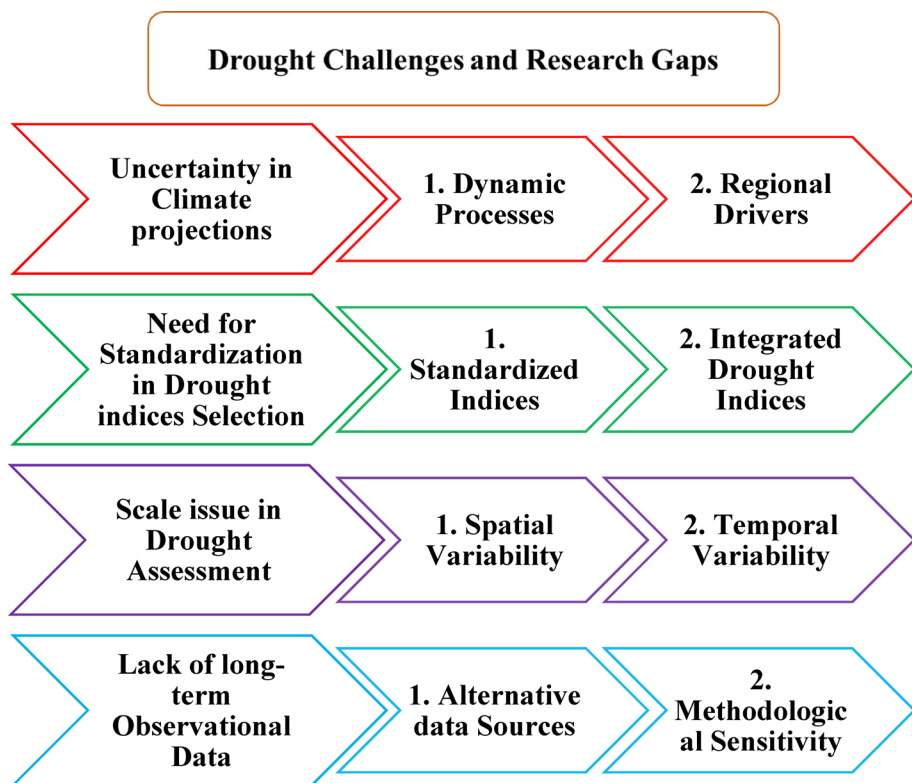


Fig. 9 Drought challenges and research gaps

development, as well as how these factors may evolve under future climate conditions (Barriopedro et al. 2023). Additionally, there are difficulties in quantifying the interactions between different processes that influence drought occurrence and their responses to climate change (Barriopedro et al. 2023). The complexity of drought as a natural hazard, with its recurrent and multifaceted nature, further complicates predictions (Aldunce et al. 2017). In order to address these challenges and research gaps, several approaches have been proposed. Approaches to reducing uncertainty in climate model projections include utilising emergent constraints (Petrova et al. 2024), developing continuous stochastic or non-stationary models (Brunner et al. 2021), and coupling regional climate models with physically based hydrological models to improve future water supply projections (Trinh et al. 2017). Additionally, integrating multiple strategies—such as agronomic and biotechnological methods alongside advanced genome editing techniques—has been proposed to enhance crop drought tolerance and mitigate the effects of more frequent and intense droughts (Nadeem et al. 2019).

7.2 Scale issues in drought assessment

Drought assessment faces several challenges and research gaps related to scale issues: Spatial and temporal variability pose significant challenges in drought monitoring and evaluation (Hasan et al. 2019). The heterogeneous nature of droughts across different spatial

scales and time periods makes it challenging to capture drought characteristics accurately. This variability necessitates a holistic approach and international collaborations to improve drought assessment methodologies (Hasan et al. 2019). One major issue is the use of non-standardised methodologies and distinct sample sizes in drought research, leading to data summary challenges and unrealistic comparisons (Hasan et al. 2019). This lack of standardisation hinders comprehensive analyses across different regions and time scales. Remote sensing techniques offer promising solutions for addressing scale issues in drought assessment. Despite advancements, challenges persist in utilising satellite data to monitor and assess the impacts of drought on forest ecosystems (Zhang et al. 2013). Enhancing data fusion techniques that integrate in situ and remotely sensed data is essential for improving drought assessment accuracy across different spatial and temporal scales. Applying various drought indices over multiple time frames reveals distinct patterns. For example, in Greece, the Standardised Precipitation Index (SPI) indicates that mild and moderate drought occurrences decrease from north to south and west to east at both 3- and 6-month scales. In contrast, severe droughts display different spatial distributions over 12 months (Livada and Assimakopoulos 2007). It underscores the necessity of incorporating multiple time scales in drought evaluations to capture a comprehensive picture of drought impacts. Addressing scale-related challenges in drought assessment requires standardised methodologies, improved data integration techniques, and consideration of spatial and temporal variability. Future research should focus on developing more robust approaches to manage drought variability across different scales, leading to more precise and comprehensive drought assessments.

7.3 Lack of long-term observational data in some regions

The frequency and intensity of droughts present several challenges and research gaps. One major challenge is the lack of long-term observational data in some regions, which hinders accurate drought assessment and trend analysis. It is evident in studies that rely on alternative or shorter data sources. For instance, Mianabadi et al. (2022) utilised CHIRPS satellite data as an alternative to ground observations for analysing drought trends in southeast Iran from 1981 to 2020 (Mianabadi et al. 2022). Similarly, Kuśmierk-Tomaszewska and Żarski (2021) analysed drought trends in central Poland, but their study was limited to the period from 1961 to 2020. The lack of extensive historical records makes it challenging to establish long-term drought patterns and distinguish natural variability from the impacts of climate change. Interestingly, different drought indices and methodologies can lead to contradictory results, even for the same region. Wang et al. (2018) demonstrate this by comparing two versions of the Palmer Drought Severity Index in China, which showed exaggerated drought severity in some areas depending on the method used for potential evapotranspiration. It highlights the need for careful selection and interpretation of drought metrics. In conclusion, while numerous studies have investigated drought trends, the lack of long-term data in many regions remains a significant obstacle. Additionally, the sensitivity of results to methodological choices highlights the importance of employing multiple approaches and indices to comprehensively understand drought patterns and their changes over time.

7.4 Need for standardisation in drought index selection and application

Standardising the selection and application of drought indices ensures accurate and consistent drought monitoring. Given the wide range of available drought indices and their

varying sensitivities to input data and climatic conditions, standardisation helps improve the reliability of these indices. Indices such as the Standardised Precipitation Index (SPI) and the Standardised Palmer Drought Index (SPDI) offer advantages in terms of spatial and temporal comparability, enabling multi-scalar calculations and facilitating the assessment of hydro-meteorological conditions across different regions (Ma et al. 2014; Liu et al. 2016).

The SPDI, in particular, strikes a balance between ease of calculation and comprehensive evaluation of water supply and demand, making it a valuable tool for drought monitoring (Ma et al. 2014). Research suggests that some standardised indices, including SPDI, demonstrate lower sensitivity to data selection than alternatives like the Self-Calibrating Palmer Drought Severity Index (SC-PDSI), highlighting their robustness in varying climatic conditions (Liu et al. 2016). It highlights the importance of understanding the sensitivity of different indices to input data and climatic conditions when selecting an appropriate drought index for a specific application. In conclusion, standardisation in drought index selection and application is essential for ensuring consistent and comparable drought assessments across different regions and time scales. The development of integrated and multi-scalar drought indices, such as the Hybrid Drought Index (HDI) and the Combined Drought Indicator (CDI), further underscores the need for standardisation to effectively combine multiple drought indicators and enhance overall drought monitoring capabilities (Mohammad et al. 2009; Cammalleri et al. 2021).

8 Conclusion

8.1 Summary of key findings

- Drought lacks a universal definition, with flash droughts emerging as a research focus since 2002. A holistic approach that incorporates climatic, remote sensing, and composite indices is essential for an accurate assessment.
- Though regional variations exist, climate change intensifies droughts, making them more frequent, severe, and prolonged. Rising global temperatures affect precipitation patterns, leading to more extreme drought conditions.
- Various drought indices, including SPI, SPEI, and PDSI, are utilised for different purposes. Remote sensing and multi-variable indices improve monitoring accuracy. The shift toward satellite-based indices highlights advancements in agricultural drought assessments.
- AI and ML techniques enhance drought prediction by analysing multiple climate variables. However, challenges like data quality, model transparency, and user trust must be addressed for effective implementation.
- High-resolution datasets (e.g., daily SPI/SPEI and weekly drought indices) enhance the precision of monitoring. Integrating multiple indices improves drought assessments, particularly for agricultural applications.
- Drought frequency and intensity are expected to rise, with agricultural and hydrological droughts becoming more prominent. Multi-index assessments and advanced climate models are necessary for accurate forecasting and resource management.

Overall, advancements in drought indices, AI integration, and high-resolution monitoring enhance drought prediction and mitigation, supporting climate adaptation and water resource planning.

8.2 Implications for drought management and policy

Drought management and policy implications are multifaceted, necessitating a comprehensive approach that considers various factors and strategies. Research indicates that climate change is expected to intensify drought conditions, particularly in Mediterranean and arid regions, highlighting the need for proactive and adaptive management strategies (Iglesias et al. 2011; Hervás-Gómez and Delgado-Ramos 2019). Effective drought management requires anticipatory modelling and long-term planning to mitigate risks effectively. For example, WaterSim 5, a predictive model for water policy and planning, has demonstrated that conventional management approaches may be unsustainable, particularly during prolonged mega-droughts. Combining various drought management policies, such as population growth management, water conservation, and water augmentation, can lead to more sustainable outcomes (Gober et al. 2016). Implementing Drought Management Plans (DMPs) in Spain has been an effective strategy for early drought warning and impact mitigation (Estrela and Vargas 2012). Studies using cooperative game theory frameworks suggest that stakeholder collaboration can significantly lower drought-related economic losses. However, during extreme water scarcity, regulatory intervention by public agencies may be necessary to ensure the protection of ecosystems while maintaining economic stability (Kahil et al. 2016). It highlights the importance of balancing various interests and priorities in drought management policies. In conclusion, effective drought management and policy require anticipatory planning, stakeholder cooperation, and adaptive strategies. Emphasis should be placed on community-based preparedness planning, improving drought prediction methodologies, and establishing robust monitoring and early warning systems (Prabhakar and Shaw 2008). Additionally, integrating ecosystem services into drought management strategies can contribute to long-term sustainability and resilience (Orimoloye et al. 2021). As climate change continues to impact water resources, policymakers must prioritise socially and economically sensitive policies that address the unique challenges faced by different regions (Iglesias et al. 2011; Bandyopadhyay et al. 2020).

8.3 Recommendations for future research

Future research on the impacts of climate change on drought should emphasise a multi-index approach for a more comprehensive assessment. Integrating meteorological, agricultural, and hydrological drought indices will enhance the ability to capture diverse drought characteristics and improve monitoring accuracy (Leng et al. 2015; Kang and Sridhar 2017).

Studies should also focus on quantifying the influence of anthropogenic factors, such as greenhouse gases and aerosols, on drought patterns. Research by Chiang et al. (2021) emphasises the importance of isolating these specific forcings to understand their role in intensifying drought.

There is a need for more regional and national studies, particularly in understudied areas such as Europe. Padrón-Monedero et al. (2024) emphasise the lack of research linking

drought to mental health outcomes in Europe, calling for studies considering environmental confounders and diverse temporal drought indices.

Given the discrepancies between drought metrics, future research should carefully evaluate the reliability of different indices. For instance, Hoerling et al. (2012) found that the Palmer Drought Severity Index (PDSI) may exaggerate future surface water deficits compared to soil moisture indicators. Studies should address these inconsistencies and ensure comparability when using different indices (Vicente-Serrano et al. 2020).

In summary, future research should:

- (1) Utilise multiple drought indices to conduct a comprehensive assessment.
- (2) Investigate specific anthropogenic influences on drought characteristics.
- (3) Expand regional studies, particularly in Europe and other understudied areas.
- (4) Address discrepancies between drought indices and ensure comparability.

These approaches will enhance the accuracy of climate change impact assessments on drought frequency and intensity.

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Declarations

Conflict of interest The authors have no conflict of interest.

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